Chapter 4 exercises

Use real Euclidean inner products unless stated otherwise. Specify the projected subspace, basis, rank assumptions, and residual.

CH04-E1

Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

1. Find a basis of $N(A)$.
2. Verify that a row-space basis is orthogonal to the nullspace.
3. Find $N(A^T)$ and explain why $C(A) = \mathbb{R}^2$.

CH04-E2

Let $a = (1, 1, 1)^T$ and $b = (2, 1, 4)^T$.

1. Find the projection p of b onto $L = \text{span}\{a\}$.
2. Find $e = b - p$.
3. Check $a^T e = 0$.
4. Give P and verify $P = P^T = P^2$.

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CH04-E3

Fit $y = c + mt$ to the points

$$(t, y) = (0, 1), (1, 2), (2, 2).$$

1. Give the design matrix A and parameter vector x.
2. Form and solve the normal equations.
3. Find p, e and check $A^T e = 0$.
4. Find the minimum sum of squared residuals.

CH04-E4

Let $A \in \mathbb{R}^{m \times n}$ have independent columns and define

$$P = A(A^T A)^{-1} A^T.$$

1. Prove $P^T = P$ and $P^2 = P$.
2. Prove $I - P$ is a projector and $P(I - P) = 0$.
3. Prove $A^T A$ is invertible.
4. Conversely, prove dependent columns make $A^T A$ singular.

CH04-E5

For

$$a_1 = (1, 1, 0)^T, \qquad a_2 = (1, 0, 1)^T$$

perform Gram-Schmidt.

1. Find orthonormal q_1, q_2.
2. Give R in $A = [a_1 \; a_2] = QR$.
3. Check $Q^T Q = I_2$ and $QR = A$.

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CH04-E6

Use A from CH04-E5 to solve $Ax = b$ by least squares, with

$$b = (1, 2, 0)^T.$$

1. Compute $Q^T b$ explicitly and solve $Rx = Q^T b$.
2. Find $p = Ax$, $e = b - p$, and $\|e\|^2$.
3. Check $A^T e = 0$ column by column without first assuming the normal equations.

CH04-E7

Let

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

1. Determine whether the columns are independent.
2. Project b onto $C(A)$.
3. Find every least-squares solution $x^\star$.
4. Explain why projection is unique but parameters are not.

CH04-E8

Let

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

1. Prove $P^2 = P$.
2. Prove $P \neq P^T$.
3. Find the range and kernel of P.
4. Explain why idempotence does not imply orthogonal projection, identifying the failure of orthogonality between the range and kernel.

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CH04-E9

For $A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \end{bmatrix}$ and $b = (3, 1)^T$, find bases and dimensions for all four fundamental subspaces. Compute the orthogonal projection p and residual e. Prove the minimum residual norm using a Pythagorean identity and find all minimizing parameters x. Prove which parameter has minimum norm. Finally decompose the particular solution $(2, -1, 2)^T$ into row-space and nullspace parts and compare its norm with that minimum.

CH04-E10

In exact arithmetic, fit $y = c + mt$ to $(0, 2), (1, 0), (2, 1)$.

1. Write the design matrix and normal equations. Show elimination steps for the intercept and slope.
2. Tabulate predictions, residuals and their squares; check $A^T e = 0$ and draw the vertical residuals in the observation plot.
3. Derive the exact increase in squared residual norm under an arbitrary parameter change $(\Delta c, \Delta m)$. Prove uniqueness. Why is a zero residual sum insufficient?
4. Center time using $s = t - 1$, then independently check the parameters and minimum residual with thin QR and $q_3 = (1, -2, 1)^T/\sqrt{6}$. Distinguish vertical direction in the observation plot from residual orthogonality in $\mathbb{R}^3$.

CH04-E11

Use $\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$ on $C[0, 1]$. Find the orthogonal projection of x^2 onto $\mathrm{span}\{1, x\}$, verify that the residual is orthogonal to both basis functions, and compute the minimum integrated squared error.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 4.

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